Product and Quotient Laws for Integer Exponents
Answer Key

Grade 8 / Pre-Algebra

Quick Answers

1. (a) the single exponent = 7, (b) the value = 128
2. x^8
3. y^5
4. (a) the single exponent = 2, (b) the value = 25
5. m^4
6. $1\frac{1}{t^6}$
7. $10x^7$
8. $3n^3$
9. =
10. (a) the corrected simplification = x^7, —

What is verified

9 of 10 problems fully machine-checked against SymPy.
— marks an answer only you can judge — problem 10. The worked solution states what a correct response must contain.

Curriculum

Level: Grade 8 / Pre-Algebra
8.EE.A.1 – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10*
Difficulty 1–5 of 5 across 13 tagged checks.

1. Expanded-form model on $2^3 \cdot 2^4$: (a) the single exponent, (b) the value.

Solution (a): Expanded, the first power is three factors of the base and the second is four more, written end to end:

$$(2 \cdot 2 \cdot 2)(2 \cdot 2 \cdot 2 \cdot 2)$$

Counting the run gives $3 + 4$ factors. $\boxed{7}$

Solution (b): Multiplying that run out doubles seven times. $\boxed{128}$

2. Simplify to a single power: $x^5 \cdot x^3$.

Solution: Same base, multiplication, so the counts of factors add: $5 + 3$. $\boxed{x^8}$

3. Simplify to a single power: $\dfrac{y^9}{y^4}$.

Solution: Same base, division, so the bottom count cancels that many factors from the top: $9 - 4$. $\boxed{y^5}$

4. Expanded-form model on $\frac{5^6}{5^4}$: (a) the single exponent, (b) the value.

Solution (a): Six factors of the base on top, four on the bottom; each bottom factor cancels one on top, leaving $6 - 4$ of them. $\boxed{2}$

Solution (b): Two factors of five multiplied together. $\boxed{25}$

5. Simplify to a single power: $m^7 \cdot m^{-3}$.

Solution: The product law holds for negative exponents as well, so add the counts: $7 + (-3)$. A negative exponent cancels factors rather than adding them, which is why the result is smaller than the first power. $\boxed{m^4}$

6. Simplify with a positive exponent: $\frac{t^4}{t^{10}}$.

Solution: Subtracting gives $4 - 10 = -6$, so the answer is t^{-6}. There are more factors on the bottom than on top, so the leftovers stay in the denominator: $\boxed{\frac{1}{t^6}}$

7. Simplify: $(2x^3)(5x^4)$.

Solution: Multiplication can be reordered, so group the number factors and the powers: $(2 \cdot 5)$ and $(x^3 \cdot x^4)$. The numbers multiply to give ten and the exponents add to give seven. $\boxed{10x^7}$

8. Simplify: $\frac{12n^8}{4n^5}$.

Solution: Divide the number factors and apply the quotient law to the powers: $12 \div 4$ and n^{8-5}. $\boxed{3n^3}$

9. Write $<$, $>$, or $=$ between $2^3 \cdot 2^5$ and $\frac{2^{10}}{2^2}$.

Solution: Use the laws rather than the arithmetic. The left side adds its exponents, $3 + 5$; the right side subtracts, $10 - 2$. Both come to the same single power of the same base, so the two quantities are equal — worth noticing, because the two laws are the same counting argument run in opposite directions. $\boxed{=}$

10. Challenge. Jonah writes $x^4 \cdot x^3 = x^{12}$. (a) Correct answer? (b) Why add, not multiply?

Solution (a): Same base and a multiplication, so the exponents add: $4 + 3$. $\boxed{x^7}$

Solution (b) — instructor-judged. Jonah multiplied the exponents. Expanded form settles it: four factors of the base written next to three more is a run of seven factors, not twelve. A numerical counterexample does the same job — with a base of two the correct product is one hundred twenty-eight, while multiplying the exponents would claim a number in the thousands. Full credit requires the expanded-form reason or an equivalent counterexample, not just the statement of the rule.